



2026

BIOLOGY

Grade 9 NCE | **Paper 3**

Time Allowed: 45 minutes **Total Marks:** 50

Topics covered: Genetics • Respiratory System • Excretory System

Candidate Name: _____

Index Number: _____

Class / Form: _____

Instructions to Candidates:

1. Write your name, index number and class in the spaces provided above.
2. Answer **ALL** questions in **ALL** sections.
3. For Section A, tick the box next to the letter of the best answer.
4. Write your answers for Sections B and C in the answer spaces provided.
5. Show all working where required — method marks may be awarded.
6. Do not use correction fluid; cross out errors neatly and rewrite.
7. The mark scheme is included at the end for self-assessment after attempting the paper.

Section A: 15 marks | **Section B:** 20 marks | **Section C:** 15 marks | **Total:** 50 marks

SECTION A — Multiple Choice Questions (15 marks — 1 mark each)

For each question, tick the box next to the letter of the **ONE** best answer.

1. The basic unit of heredity is called a: [1]
- ☐ A) Chromosome
- ☐ B) Gene
- ☐ C) Nucleus
- ☐ D) Cell
2. Human body cells contain how many chromosomes? [1]
- ☐ A) 23
- ☐ B) 46
- ☐ C) 48
- ☐ D) 92
3. In Mendel's experiment, a tall plant (TT) is crossed with a short plant (tt). What proportion of offspring are tall? [1]
- ☐ A) 0%
- ☐ B) 25%
- ☐ C) 50%
- ☐ D) 100%
4. An organism that carries one dominant and one recessive allele for a trait is said to be: [1]
- ☐ A) Homozygous dominant
- ☐ B) Homozygous recessive
- ☐ C) Heterozygous
- ☐ D) Codominant
5. Which molecule carries the genetic code in chromosomes? [1]
- ☐ A) RNA
- ☐ B) ATP
- ☐ C) DNA
- ☐ D) Protein
6. A characteristic that is expressed even when only one copy of the allele is present is called: [1]
- ☐ A) Recessive
- ☐ B) Dominant
- ☐ C) Co-dominant

☐ D) Homozygous

7. The alveoli in the lungs are adapted for efficient gas exchange by:

[1]

- ☐ A) Having thick walls
- ☐ B) Being surrounded by cartilage
- ☐ C) Having a large surface area and rich blood supply
- ☐ D) Secreting mucus

8. During inhalation, the diaphragm:

[1]

- ☐ A) Relaxes and moves upward
- ☐ B) Contracts and moves downward
- ☐ C) Moves to the left
- ☐ D) Remains stationary

9. Which of the following is the correct order of the respiratory tract?

[1]

- ☐ A) Nose → Trachea → Bronchus → Bronchioles → Alveoli
- ☐ B) Alveoli → Bronchioles → Trachea → Nose
- ☐ C) Nose → Alveoli → Bronchus → Trachea
- ☐ D) Trachea → Nose → Alveoli → Bronchioles

10. The main excretory organ for filtering blood and producing urine is the:

[1]

- ☐ A) Liver
- ☐ B) Lungs
- ☐ C) Skin
- ☐ D) Kidney

11. The functional unit of the kidney is called the:

[1]

- ☐ A) Ureter
- ☐ B) Nephron
- ☐ C) Urethra
- ☐ D) Glomerulus

12. Which waste product results from the breakdown of amino acids in the liver?

[1]

- ☐ A) Carbon dioxide
- ☐ B) Glucose
- ☐ C) Urea
- ☐ D) Lactic acid

13. During filtration in the kidney, which of these molecules is **NOT** filtered out of the blood?

[1]

- ☐ A) Glucose

- ☐ B) Water
- ☐ C) Red blood cells
- ☐ D) Urea

14. In a monohybrid cross between two heterozygous plants ($Rr \times Rr$), what is the expected phenotype ratio? [1]

- ☐ A) 1:1
- ☐ B) 2:1
- ☐ C) 3:1
- ☐ D) 4:1

15. Which gas makes up the highest percentage of exhaled air, other than nitrogen? [1]

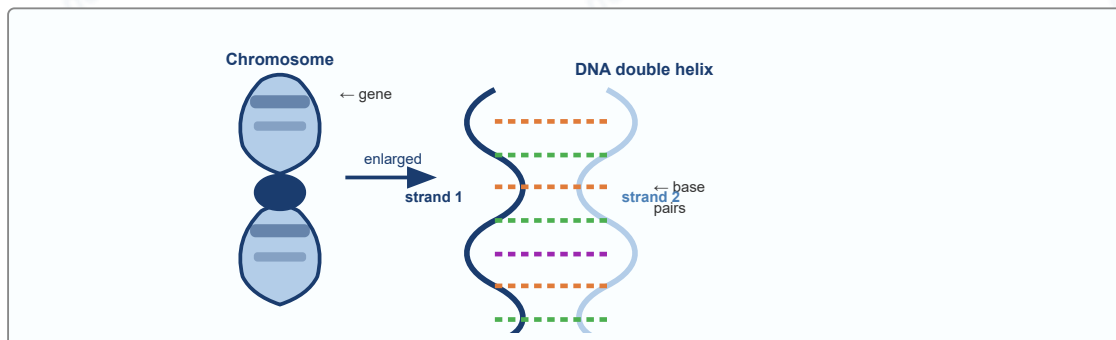
- ☐ A) Oxygen
- ☐ B) Carbon dioxide
- ☐ C) Water vapour
- ☐ D) Argon

SECTION B — Structured Questions (20 marks — 4 questions × 5 marks each)

Answer **ALL** questions. Write your answers in the spaces provided.

B1 — Chromosomes and DNA [5 marks]

Study the diagram below showing the relationship between a chromosome, DNA and genes.



- (a) State the relationship between genes, chromosomes and DNA.

[2]

- (b) Human gametes contain 23 chromosomes. Explain why this is important for sexual reproduction.

[2]

- (c) What is meant by a 'mutation'? Give ONE example.

[1]

B2 — Mendelian Genetics [5 marks]

In guinea pigs, black coat (**B**) is dominant over white coat (**b**). Two heterozygous guinea pigs are crossed.

- (a) Write the genotypes of the two parent guinea pigs.

[1]

- (b) Complete the Punnett square below to show the possible offspring genotypes.

[2]

	B	b
B		
b		

- (c) State the phenotype ratio of the offspring.

[1]

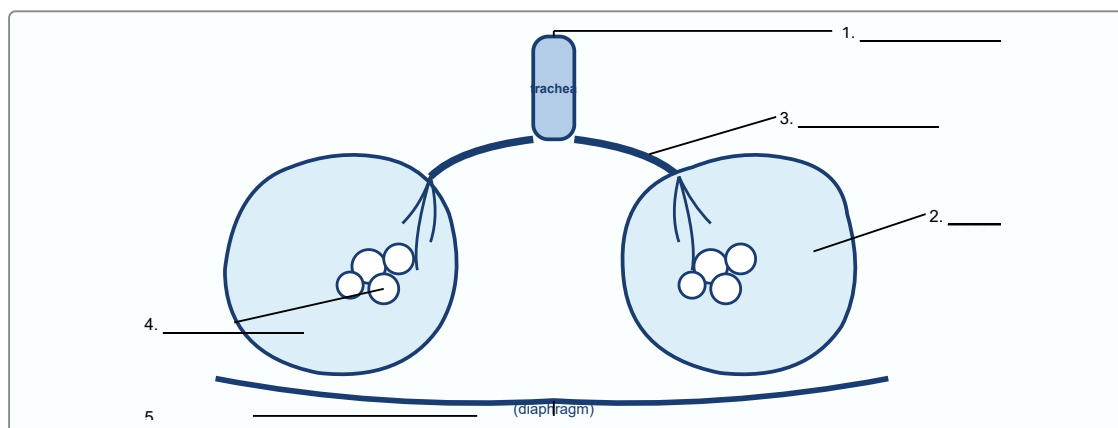
- (d) What is the probability of an offspring having a white coat?

[1]

B3 — The Respiratory System [5 marks]

Study the diagram of the human respiratory system below. Label the five structures indicated by the arrows.

[5]



- (a) Describe what happens to the volume of the chest cavity during inhalation.

[1]

- (b) Explain how the alveoli are adapted for efficient gas exchange. Give **THREE** adaptations.

[3]

- (c) Name ONE substance that damages the alveoli and state how it does so.

[1]

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B4 — The Excretory System [5 marks]

- (a) Name THREE waste products removed from the body and state the organ responsible for each.

[3]

Waste product	Organ responsible

- (b) Explain the process of ultrafiltration in the kidney, using the terms *glomerulus* and *Bowman's capsule*.

[2]

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SECTION C — Extended Questions (15 marks)

Answer **ALL** questions. Write in full sentences where appropriate.

C1 — Genetics [8 marks]

In humans, tongue rolling (**R**) is dominant over non-rolling (**r**). A tongue roller who is heterozygous marries a non-tongue roller.

(a)

(i) State the genotypes of both parents.

[1]

(ii) Draw a Punnett square for this cross.

[2]

(iii) What percentage of children would be expected to roll their tongue?

[1]

(iv) If they have 4 children, predict how many would be tongue rollers.

[1]

(b) Explain the difference between genotype and phenotype. Give one example.

[2]

(c) Explain why children from the same parents are not identical to each other (except identical twins).

[1]

C2 — Respiratory and Excretory Systems [7 marks]

- (a) Describe the path of air from the nose to the alveoli, mentioning at least **FOUR** structures. [2]

- (b) Explain what happens to glucose and urea during filtration and reabsorption in the kidney. [2]

- (c) Smoking damages cilia in the respiratory tract. Explain what effect this has on a person's health. [2]

- (d) Name **ONE** other organ involved in excretion (not the kidneys) and state what waste product it removes. [1]

MARK SCHEME — Paper 3 (Biology)

For teacher/tutor use only. Accept any equivalent correct biological answer. Points marked with 'OR' are acceptable alternatives. Do not deduct marks for minor spelling errors if the meaning is clear.

SECTION A — Multiple Choice Answers

Q	Answer	Q	Answer	Q	Answer
1	B — Gene	2	B — 46	3	D — 100%
4	C — Heterozygous	5	C — DNA	6	B — Dominant
7	C — Large surface area & rich blood supply	8	B — Contracts and moves downward	9	A — Nose→Trachea→Bronchus→Bronchioles→Alveoli
10	D — Kidney	11	B — Nephron	12	C — Urea
13	C — Red blood cells	14	C — 3:1	15	A — Oxygen

SECTION B — Structured Questions

B1 — Chromosomes and DNA		
Part	Accepted Answer	Marks
(a)	Chromosomes are found in the nucleus of cells and are made of DNA (1). Genes are specific sections/segments of DNA that code for a particular characteristic or protein (1).	2
(b)	During fertilisation, a gamete from each parent fuses together (1). If each gamete had 46 chromosomes, the resulting zygote would have 92, which would be fatal/incompatible with life; having 23 ensures the zygote receives the correct number of 46 chromosomes (1).	2
(c)	A mutation is a random/spontaneous change in the DNA/genetic material of an organism (1). Example: sickle cell anaemia / Down syndrome / albinism / any other valid mutation (1) — only 1 mark total for this part.	1

B2 — Mendelian Genetics		
Part	Accepted Answer	Marks
(a)	Both parents: Bb (heterozygous) — must show both parents as Bb.	1
(b)	Punnett square correctly completed: BB, Bb, Bb, bb (1 mark for correct parental gametes shown; 1 mark for all four offspring genotypes correct).	2
(c)	3 black : 1 white (OR 3:1 black to white).	1
(d)	1 in 4 OR 25% OR 0.25.	1

B3 — The Respiratory System		
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Part	Accepted Answer	Marks
Labels	1 = Trachea 2 = Right lung (OR Left lung) 3 = Bronchus 4 = Alveoli 5 = Diaphragm. (Accept bronchi for 3; accept alveolus for 4.) [1 mark per correct label, max 5 — but marks for labels are in the diagram instructions, not a separate sub-question here — award up to 2 marks for correct identification of any 2 structures if used as a labelling task; otherwise treat labels as informational.]	—
(a)	The volume of the chest cavity increases/gets bigger (1).	1
(b)	Any THREE of: (1) Large surface area — many alveoli provide a huge total surface area for diffusion. (2) Very thin walls (one cell thick) — short diffusion distance. (3) Rich/dense blood supply (capillary network) — maintains a steep concentration gradient. (4) Moist lining — gases dissolve for transport. (5) Good ventilation — keeps concentration gradient steep.	3
(c)	Tar (in cigarette smoke) — destroys the walls of alveoli, reducing surface area for gas exchange, leading to emphysema. OR: Particulates cause inflammation and scarring. [1 mark for substance + 1 mark for mechanism — only 1 mark available for this part, so accept substance + brief how.]	1

B4 — The Excretory System		
Part	Accepted Answer	Marks
(a)	Any THREE correct pairs (1 mark each): Urea → Kidneys / Carbon dioxide → Lungs / Water → Kidneys, Lungs, or Skin / Bile pigments → Liver (excreted via gut) / Excess salts → Kidneys or Skin.	3
(b)	Blood enters the glomerulus (a knot of capillaries) at high pressure (1). Small molecules — water, glucose, urea, salts — are forced out of the blood into the Bowman's capsule by the pressure; large molecules and blood cells remain in the blood (1).	2

SECTION C — Extended Questions

C1 — Genetics [8 marks]		
Part	Accepted Answer	Marks
(a) (i)	Tongue roller (heterozygous): Rr Non-tongue roller: rr.	1
(a) (ii)	Punnett square: Parental gametes R and r from Rr parent; r and r from rr parent. Offspring: Rr, Rr, rr, rr. (1 mark for correct gametes labelled; 1 mark for all four offspring correct.)	2
(a) (iii)	50% (OR 2 out of 4 / 1 in 2 / 0.5).	1
(a) (iv)	2 children (out of 4) would be expected to be tongue rollers.	1
(b)	Genotype is the genetic make-up / alleles an organism possesses (e.g. Rr) (1). Phenotype is the observable/physical characteristic expressed (e.g. tongue roller) (1).	2
(c)	During sexual reproduction, gametes are produced by meiosis, which involves random assortment/shuffling of chromosomes (1); fertilisation then combines gametes randomly, so each child receives a different combination of alleles from their parents (accept: genetic variation through independent assortment and random fertilisation).	1

C2 — Respiratory and Excretory Systems [7 marks]
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Part	Accepted Answer	Marks
(a)	Nose (nasal cavity) → Pharynx / throat → Larynx / voice box → Trachea → Bronchi → Bronchioles → Alveoli. Must include at least 4 named structures in correct order for 2 marks; 1 mark for 2–3 correct structures in order.	2
(b)	During filtration: both glucose and urea (along with water and salts) are filtered out of the blood into the Bowman's capsule / filtrate (1). During reabsorption: glucose is completely reabsorbed back into the blood by active transport in the tubules; urea is NOT reabsorbed — it remains in the filtrate and is excreted as urine (1).	2
(c)	Cilia normally sweep mucus (and trapped dust/bacteria) upwards towards the throat to be swallowed or expelled (1). When cilia are damaged, mucus and pathogens/bacteria accumulate in the airways, increasing the risk of infection / bronchitis / smoker's cough (1).	2
(d)	Any ONE correct: Lungs — carbon dioxide / Liver — bile pigments (bilirubin) / Skin — water and salts (in sweat). (1 mark for organ + correct waste product.)	1

Mark Summary: Section A: 15 | Section B: 20 | Section C: 15 | **Total: 50 marks**